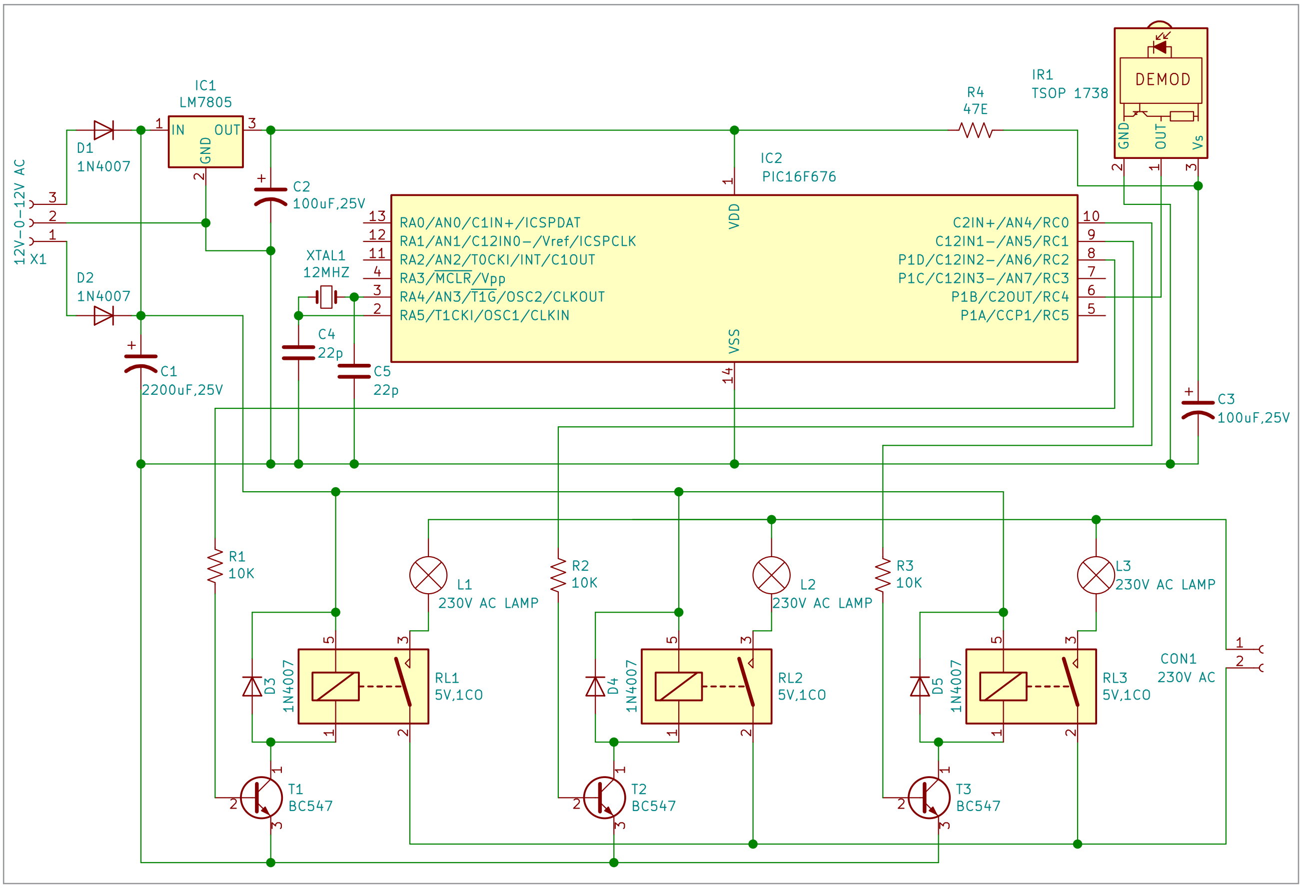

Fig. 7: Circuit diagram

The working of this project is fairly simple. When a button is pressed on the IR remote, it sends a sequence of codes in the form of encoded pulses using 38kHz modulating frequency. These pulses are received by the TSOP 1738 sensor and then read by the microcontroller. The microcontroller decodes the received train of the pulses into a hex value and compares it with the predefined hex values in the program.

Whenever a key is pressed on the IR remote, the transmitted IR signal is received by an IR receiver and the key is decoded by the software. In this project, the entire 32-bit string (4 bytes of NEC protocol) is used for decoding. In this 32-bit string, the third byte is compared in the program for controlling the lights/fans. If any match occurs, the controller performs a relative operation by triggering the respective relay through transistor BC547 and the corresponding result is indicated by an on-board LED. The three LEDs in the circuit show the status of the relays.

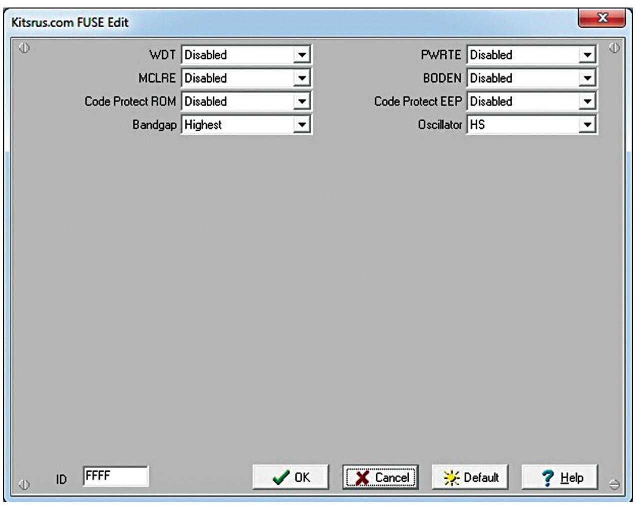

Fig. 8: Fuse bit settings

In this project, key numbers 2, 4, and 6 of the IR remote have been used for controlling the three relays. Key 2 toggles relay RL1, key 4 toggles relay RL2, and key 6 toggles relay RL3.

Software

The circuit uses the software program loaded into the internal memory of PIC16F676. The program is compiled using Mikro C PRO version 7.2.0 for PIC compiler and uploaded into PIC16F676. The software main.c is written in embedded C language, which allows writing